# GenAI for Engineers Lesson Plan

*IES | v3.4 | LP-aligned one-day delivery*

Follow the supplied Lesson Plan with Hands-On Labs (v2 - PUB Feedback) module order and clock times. Retain its six modules; consolidate learner work into only 65 minutes of guided Excel exercises within Module 3. Other practical examples are trainer-led demonstrations or discussion. LO4 includes a guided paired Copilot Agent build, two preview tests and approved training deployment evidence.

09:00–17:00 = 480 minutes: 390 teaching minutes + 60 lunch + 30 tea. The source LP calls this seven training hours excluding lunch; that includes tea. This delivery plan uses 6.5 net teaching hours. Guided Excel time is 20 + 20 + 25 = 65 minutes, included within Module 3, never added on top.

| Time | LP module | Delivery allocation |
|---|---|---|
| 09:00–10:00 | 1 · Copilot ecosystem | 15 overview; 10 access check; 20 Excel demonstration; 15 discussion |
| 10:00–11:15 | 2 · Prompting for engineering documents | 20 RISE/E-MAP; 20 document demonstration; 20 constraints and persona review; 15 prompt critique |
| 11:15–11:30 | Tea break | 15 minutes |
| 11:30–13:00 | 3 · Engineering methodology and Excel | 15 primer; 20 guided FMEA; 20 guided DMAIC; 25 guided SPC; 10 shared review |
| 13:00–14:00 | Lunch | 60 minutes |
| 14:00–15:15 | 4 · Copilot and reasoning agents | 15 concepts; 25 FMEA/report assistant demo; 20 research/analysis comparison; 15 questions and buffer |
| 15:15–15:30 | Tea break | 15 minutes |
| 15:30–16:15 | 5 · SharePoint knowledge base and grounded agent | 5 explain; 7 inspect site/sources; 10 agent; 10 tests; 8 citation/access; 5 evidence |
| 16:15–16:45 | 6 · Apply the 7-Step AI Security Framework | 20 applied seven-step review and release decision; 10 personal prompt toolkit |
| 16:45–17:00 | Closing and Q&A | 10 recap; 5 next action |

## Only three short guided exercises

Use the separate Starter.xlsx in each lab. Each workbook contains one small source table and one prompt. Open Worked-Reference.xlsx only after the learner has checked the Copilot draft.

## 1 · Copilot ecosystem

Show the same engineering question in Chat and Excel. Identify which evidence is available in each surface. Complete the access check once; use a paired screen or the supplied worked reference when access cannot be resolved within ten minutes.

## 2 · Prompting for engineering documents

Teach RISE as the main LO3 framework. R - Role: name the perspective without granting authority. I - Input: name the exact table, document, revision and boundaries. S - Steps: state the calculations, comparisons and checks in order. E - Expectations: specify the output format, evidence, constraints and

acceptance test. Demonstrate vague prompt -> RISE prompt -> checked output -> one refinement. Learners apply the same structure to the FMEA, DMAIC and SPC prompts in Module 3.

LO3 evidence: learners label R, I, S and E in one engineering prompt, use it in Copilot, verify one result, and refine one instruction when the draft is incomplete.

## 4 · Copilot and reasoning agents

8 min: explain the six-step FMEA-to-report workflow. 15 min: pairs create the IES Pump FMEA Assistant and add the instruction/template. 12 min: run the complete FM01 and missing-Detection tests; refine one instruction. 8 min: human approval and report draft. 12 min: create, publish or share to the approved training audience and record status. 20 min LO5 guided activity: Researcher cited comparison 8 min, Analyst DMAIC analysis 8 min, independent checks and debrief 4 min. Total: 75 minutes.

## LO5 guided activity - Researcher and Analyst

Learning evidence: use Researcher to create a technical comparison and open one supporting citation. Use Analyst with Lab-02-DMAIC/Starter.xlsx, then independently verify Baseline 36/400 = 9.0%, Pilot 7/200 = 3.5%, change -5.5 percentage points and relative reduction 61.1%.

Researcher 8 min: select Researcher under Agents, provide TR-401-Training-Extract.pdf and the RISE prompt, review the cited comparison and open one citation. Analyst 8 min: select Analyst, attach the DMAIC workbook, request totals, rates, change, a simple visual, a limitation and a next evidence check. Review 4 min: compare the output with the expected calculations and reject causal or sustained-improvement claims.

Resources: Trainer-Demonstrations/REASONING-AGENT-PROMPTS.md; Trainer-Demonstrations/REASONING-AGENT-CHECKS.md; https://learn.microsoft.com/en-us/microsoft-365/copilot/researcher-agent; https://support.microsoft.com/en-US/Microsoft-365-Copilot/get-started-with-analyst-in-microsoft-365-copilot

## 5 · SharePoint knowledge base and grounded agent

5 min: explain SharePoint grounding and the productivity benefit. 7 min: inspect the private IES site and four named synthetic sources. 10 min: create or inspect the folder-scoped agent. 10 min: run supported-fact and missing-evidence tests. 8 min: open one citation and check revision, scope and permissions. 5 min: record evidence and debrief. Total: 45 minutes. If learners lack edit permission, the trainer demonstrates creation and pairs evaluate the tests; record actual execution status. Prepared resource: private site https://tertiaryinfotech.sharepoint.com/sites/IES-GenAI-for-Engineers; Documents/Engineering Knowledge Base; IES Engineering Knowledge Assistant; Trainer-Demonstrations/SHAREPOINT-AGENT-INSTRUCTIONS.md and SHAREPOINT-AGENT-TESTS.md.

## 6 · Apply the 7-Step AI Security Framework

20 min guided application: use the Module 5 SharePoint-agent answer and complete all seven rows in AI-SECURITY-EVIDENCE-LOG.csv — classify the synthetic data; confirm the private site, source scope and access; inspect the secure prompt; open one citation and reproduce one calculation; record human sign-off; preserve a reconstructable audit trail; and state the containment and reporting response to a security trigger. Make an APPROVED FOR TRAINING USE, RETURN FOR CORRECTION or DO NOT SHARE decision. 10 min: adapt one reusable prompt to carry source, protection, verification and review controls. No additional lab folder.

# Module 3 · FMEA → DMAIC → SPC | 90 minutes

### 11:30–11:45 · Methodology primer

Explain FMEA and RPN; Six Sigma/DMAIC and its five phases; Pareto, fishbone/Ishikawa and 5-Whys; then SPC, common versus special cause, control charts and Cp/Cpk. Use plain language and one pump example before any Copilot task.

### 11:45–12:05 · Lab 1 FMEA

Open Labs/Lab-01-FMEA/Starter.xlsx. Preserve four supplied rows and ratings; calculate RPN, rank rows and apply the Severity 9 gate. Independent checks: FM01 = 140 and FM04 Severity = 9.

### 12:05–12:25 · Lab 2 DMAIC

Open Labs/Lab-02-DMAIC/Starter.xlsx. Calculate Baseline and Pilot deviation rates and draft five DMAIC rows. Independent checks: 9.0%, 3.5%, and no causal claim. Relate Pareto, fishbone and 5-Whys to the Analyse phase.

### 12:25–12:50 · Lab 3 SPC

Open Labs/Lab-03-SPC/Starter.xlsx. Calculate baseline limits, flag monitoring rows and draft a chart interpretation. Independent check: T10 = 59.4 L/h against the LCL. Distinguish a signal from a specification or diagnosis.

### 12:50–13:00 · Shared review

Each learner states one output, one independent check and one evidence limitation. Stop at 13:00 for lunch. If Copilot stalls for more than two minutes, inspect the Worked-Reference formula and continue the review without extending the block.

### Outcome coverage

LO1 methodology: primer and guided Excel review. LO2 documentation: trainer examples and Excel outputs; no independent Word/PPT pack. LO3 prompting: framework, critique and three selected prompts. LO4: guided paired agent build, test and training deployment. LO5: guided Researcher and Analyst activity with citation and calculation evidence. LO6: guided private SharePoint knowledge base, grounded-agent tests and citation verification. LO7: applied seven-step security record, release decision and incident-response trigger. Independent agent deployment is not a completion claim for this shortened delivery.

### Presentation teaching sequence — v3.4

Module 1 introduces AI, Copilot and the pump case. Module 2 builds clear, reviewable prompts with RISE and E-MAP. Module 3 follows FMEA → DMAIC → SPC: explain each method before its Copilot in Excel exercise. Module 4 covers the multi-step Copilot Agent and reasoning agents. Module 5 creates and grounds a SharePoint knowledge-base agent. Module 6 applies all seven AI security steps before closing.

# Using the expanded 259-slide deck

Slides 1–138 are the one-day teaching route. Slide 139 introduces optional reference material. Slides 140–259 contain 30 worked examples; they are not additional scheduled labs. Keep the original six-module timetable and the 25/20/20-minute guided Excel exercises.

Reference index: FMEA slides 140-163; DMAIC 164-187; SPC 188-219; Prompting 220-239; Agents 240-247; Grounding 248-255; Security 256-259.

Each reference example follows data, prompt, illustrative answer and verification. Supporting CSVs, prompts and Worked-Examples.pdf are in Reference-Examples. The trainer demonstrations use the supplied Trainer-Demonstrations pack.

# LO2 practical workflow: create the engineering output with Copilot 365

Repeat five actions in each exercise: open the small Starter workbook; select the named table; open Copilot in Excel; paste the supplied RISE prompt; inspect the output and independently verify one result. Open Worked-Reference.xlsx only after the attempt.

### Lab 1 - FMEA (20 minutes)

Source: Labs/Lab-01-FMEA/Starter.xlsx · named table FMEAInput.

Copilot prompt: Act as an FMEA facilitator. Use only FMEAInput and preserve every row and supplied rating. Calculate RPN = Severity × Occurrence × Detection, rank the rows, and flag Severity 9 or higher. Add one evidence request and action per row. Show the formula. Do not invent incidents, change ratings or approve the design.

Independent check: Verify FM01: 7 × 5 × 4 = 140; confirm FM04 is flagged because Severity = 9.

### Lab 2 - DMAIC (20 minutes)

Source: Labs/Lab-02-DMAIC/Starter.xlsx · named table WeeklyData.

Copilot prompt: Act as a Six Sigma project assistant. Use only WeeklyData. Calculate Baseline and Pilot deviation rates. Create a five-row Define, Measure, Analyse, Improve, Control summary. Separate facts from hypotheses, show formulas, and state that the short pilot does not prove cause.

Independent check: Verify Baseline = 36/400 = 9.0% and Pilot = 7/200 = 3.5%; do not claim causation.

### Lab 3 - SPC (25 minutes)

Source: Labs/Lab-03-SPC/Starter.xlsx · named table PumpReadings.

Copilot prompt: Act as an engineering data assistant. Use only PumpReadings. From the eight Baseline rows, calculate the mean, sample standard deviation, UCL and LCL. Flag T09–T12, create a line chart with the limits, and give a three-sentence interpretation. Show formulas. Do not claim a root cause or that the pump is safe.

Independent check: Verify T10 = 59.4 L/h against the calculated LCL; interpret a signal as a reason to investigate.

### Microsoft 365 Copilot links

Microsoft 365 Copilot: https://m365.cloud.microsoft/chat

Get started with Copilot in Excel: https://support.microsoft.com/en-us/office/get-started-with-copilot-in-excel

How Copilot Chat works in Microsoft 365 apps: https://support.microsoft.com/en-us/microsoft-365-copilot/how-copilot-chat-works-in-microsoft-365-apps